

EMADELDEEN HAMDAN

Chicago, IL

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📄 <https://scholar.google.com/citations?hl=en&user=HeFM-v0AAAAJ>

Education

University of Illinois Chicago

Jan. 2023 – Present

Ph.D., Electrical and Computer Engineering

Chicago, IL

- Researching efficient deep learning and computer vision methods for remote sensing and medical imaging applications, with a focus on lightweight AI architectures and foundation models.

University of Illinois Chicago

Sep. 2020 – May 2022

Bachelors, Electrical and Computer Engineering

Chicago, IL

- Magna Cum Laude Honors.

Professional Summary

Ph.D. Candidate in Electrical and Computer Engineering at the University of Illinois Chicago developing efficient AI and computer vision models for remote sensing and medical imaging. Research interests include efficient lightweight deep learning architectures, transfer learning, foundation models, and signal processing techniques for wildfire monitoring, MRI reconstruction, and cancer imaging. Expected Ph.D. graduation date May 2027.

Experience

Springer Nature

July 2025 – Present

Assistant Editor

Chicago, IL

- Reviewed and assessed technical manuscripts in machine learning and signal processing for publication quality and clarity.
- Collaborated with editors and reviewers to ensure rigorous evaluation of research contributions and methodological soundness.

Northwestern University

May 2025 – Present

Research Assistant – Machine and Hybrid Intelligence Lab

Chicago, IL

- Conducted research in AI-driven medical imaging and prostate cancer analysis under the supervision of Prof. Ulas Bagci and Prof. Ahmet Enis Cetin.

University of Illinois Chicago

Jan 2023 – Present

Research & Teaching Assistant, Electrical and Computer Engineering

Chicago, IL

- **Research Assistant**
 - * Developed efficient and scalable machine learning models for computer vision and signal processing applications.
 - * Designed transform-based architectures (Hadamard, DCT) and state-space models to reduce computational complexity while maintaining performance.
 - * Applied models to MRI reconstruction, medical image segmentation, and large-scale environmental detection tasks.
 - * Implemented and trained models in PyTorch with GPU acceleration, optimizing performance for large datasets.
- **Teaching Assistant**
 - * Supported instruction for undergraduate and graduate courses in Digital Signal Processing and Neural Networks.
 - * Led problem-solving sessions and assisted in designing assignments to improve student understanding of core ML and signal processing concepts.

Handshake AI

Jan 2026 – May 2026

AI Trainer – Generative AI Model Evaluation

Chicago, IL

- Evaluated large-scale generative AI model outputs across multimodal tasks, improving response quality, factual accuracy, and alignment.

L3DFX LLC

May 2022 – Jan 2023

Electrical Engineer

Bolingbrook, IL

- Designed and tested signal processing circuits and embedded systems for real-world applications.
- Integrated microcontroller-based solutions for control systems, improving system reliability and performance.

Selected Publications

Hybrid Multi-Dimensional MRI Prostate Cancer Detection via Hadamard-Based Bias Correction and Residual Networks.

Designed a hybrid transform-based pipeline for MRI bias correction and classification, improving cancer detection accuracy by 30%. Accepted to *EMBC*, 2026.

Discrete Cosine Transform Based Decorrelated Attention for Vision Transformers.

We introduce two methods leveraging the Discrete Cosine Transform (DCT) to enhance the efficiency and performance of Vision Transformers. Accepted to *IJCAI*, 2026.

Zonal-Aware Prostate Cancer Detection via Hadamard-Fused Mamba-UNet.

Introduced a transform-enhanced state-space architecture combining Hadamard representations with Mamba-based models, achieving real-time prostate zonal segmentation (~ 0.01 s) and improved cancer detection performance. Submitted to *MICCAI Workshops*, 2026.

Real-time Instantaneous Phase Estimation Using a Deep Dual-Branch Complex Neural Network.

Proposed a lightweight neural architecture for real-time instantaneous phase estimation of neural oscillations, with FPGA implementation for low-latency deployment. Published in *IEEE Transactions on Biomedical Engineering (TBME)*, 2025.

Reconfigurable Electromagnetically Unclonable Functions Based on Graphene RF Modulators.

Assisted in the development of a robust hardware security primitive leveraging electromagnetic PUFs for device authentication and cryptographic identification. Published in *ACS Nano*, 2025.

Wildfire Detection via Transfer Learning: A Survey.

Surveyed transfer learning approaches for large-scale wildfire detection, highlighting challenges in generalization and dataset bias. Published in *Signal, Image and Video Processing*, 2024.

Technical Skills

Programming: Python, C, MATLAB

Machine Learning: PyTorch, TensorFlow, State-Space Models (Mamba), Transform-based Learning (Hadamard, DCT), Computer Vision, Representation Learning, Generative Models (LLMs, VLMs)

Systems & Efficiency: GPU Training (CUDA), Hardware-Aware ML, Low-Complexity Model Design, FPGA Deployment, High-Level Synthesis (HLS)

Tools: Linux (CLI), Git, VS Code

Awards and Leaderships

IEEE Communication Society

May 2025 – Present

Secretary – UIC Student Branch Chapter

Chicago, IL

- Organized technical events and workshops focused on AI, signal processing, and communications systems.
- Coordinated with researchers and industry speakers to promote knowledge exchange and student engagement.

NASA Beyond the Algorithm Challenge

Sep 2025

- Top 9 Finalist** — Developed transform-based ML layers (Hadamard, Haar wavelets) for efficient computing in quantum and in-memory systems (Team NVSense).

NASA MSI Incubator Wildfire Climate Tech Challenge

Mar 2024

- Runner-Up** — Built a computer vision system for wildfire situational awareness and early detection (Team FIRESENSE).

NSF I-Corps (POSE Program)

Feb 2024

- Entrepreneurial Lead** — Led commercialization efforts for hardware-aware ML solutions (Team HLS4ML).